

# STAT 496 Capstone

Andrew Michael Sousa, David Schutt, Sebastian Villarroel

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## 1 Introduction

Large language models (LLMs) are increasingly used in applications that require generating answers to structured questions, including tutoring systems, customer support, and automated reasoning tools. In many of these settings, users interact with models using natural language, and the phrasing of a prompt can vary widely depending on the individual. For example, two users may ask the same question using different tones, such as neutral, polite, or rude language. Understanding whether these differences in tone affect the correctness of model responses is important, as it helps determine how robust LLMs are to variation in user input.

If model performance varies depending on tone, this could introduce inconsistencies in real-world deployments. For instance, an educational tool powered by an LLM should provide accurate answers regardless of how politely or directly a student phrases a question. Similarly, automated systems used in professional or operational settings must remain reliable even when inputs are informal, abrupt, or emotionally charged. Ensuring that models behave consistently across different tones is therefore important for fairness, usability, and reliability.

A recent study by Dobariya and Kumar (2025) investigated how prompt politeness affects the accuracy of large language models using multiple-choice questions across mathematics, science, and history. The researchers created 50 base questions and rewrote each question into five tone variants (Very Polite, Polite, Neutral, Rude, and Very Rude), resulting in 250 total prompts. They evaluated ChatGPT-4o using these prompts and measured accuracy by comparing the model’s responses to the known correct answers. Their results showed that tone had a statistically significant effect on model performance. Surprisingly, rude prompts achieved the highest accuracy (84.8 percent), while very polite prompts had the lowest accuracy (80.8 percent). The authors concluded that prompt tone is an important factor affecting LLM accuracy and that models remain sensitive to linguistic phrasing even when the underlying question remains unchanged. (<https://arxiv.org/pdf/2510.04950>)

In this study, we investigate whether prompt tone affects the accuracy of LLM responses. Specifically, we compare model performance across three tones: Normal, Polite, and Rude. Using a fixed set of prompts with known answers, we measure whether response accuracy differs across tones when the underlying question remains the same. This allows us to isolate the effect of tone and evaluate whether LLMs produce consistent and reliable answers regardless of how a question is phrased.

## 2 Experimental Design

To determine how tone affects the quality of responses from an LLM, we created a variety of prompts, added text to the prompt that match the current tone, and assessed the accuracy of the following LLM response.

### 2.1 LLM Model and Answer Structuring

For this experiment, we utilized the LLM model LLama 3.2 3B Instruct, an instruction-based model trained by Meta.

To simplify the assessment of the LLama LLM, we set specific instructions for how the LLM should answer. Given that LLama is an instruction-based model, we specified that the LLM is an answer-only model and should only output the final answer without including reasoning or restating the question. The code used is provided in the Appendix. A

### 2.2 Creating LLM Prompts

An important aspect of our experiment was to see how LLMs can extract information from a given question instead of simply looking an answer up. To see if the LLM was better at answering different styles of questions, our "extracting questions" were divided into three sections:

1) Temporal: Questions involving the analysis of times and dates

*Example: "A university library opens at 8:00 AM on weekdays, opens at 10:00 AM on Saturdays, and remains closed until noon on Sundays; if today is Sunday and a student wants to enter as soon as possible, what time can they do so?"*

2) Conditional: Identifying current conditions considering specified constraints.

*Example: "A software license is valid only if it is activated within 30 days of purchase and registered to a single device; if it is activated after 45 days, is it valid?"*

3) Arithmetic Shifts: Questions involving shifting numbers and times.

*Example: "A ticket costs \$60 and receives a \$12 discount; what is the final price?"*

We generated 100 prompts for each section, for a total of 300 prompts. The full list of generated prompts can be found [here](#). Each question also has an accompanied answer.

*\*Note: Some of the questions + answers were generated by a LLM model, ChatGPT. We verified that answers given by ChatGPT were correct\**

## 2.3 Difference in Tones

In our experiment, we utilized three different tones: (1) Normal, (2) Polite, and (3) Rude. Our intention was to assess whether there was any significant difference in response accuracy across the three different tones.

*Normal Example: Instructions and question only*

*Polite Example: "Hi there, would you please help me answer the following question"*

*Rude Example: "Answer quick, sucker."*

## 2.4 Response Priority

Within each tone, we also tried 3 different phrases prioritizing something in the LLM responses to see if that made any difference in accuracy. One phrase prioritized nothing from the LLM, one prioritized ensuring answers were correct, and the last prioritized speed.

*Normal Tone, No Priority Example: Question + Prioritize Speed*

*Polite Tone, Accuracy Priority Example: "Hi there, would you please help me answer the following question. Please be as accurate as possible!"*

*Rude Tone, Speed Priority Example: "Answer quick, sucker. If you take too long I'll go to a real model like Claude."*

The different phrases are provided in the Appendix. B

Each style of tone and response priority answered the same 100 prompts within each question type. In total, we produced 2700 responses.

To clarify the structure of the experiment, a diagram is provided below:

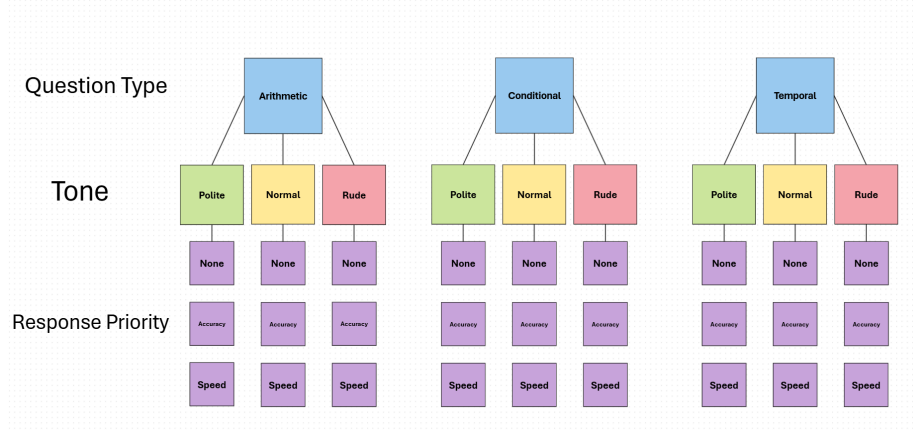


Figure 1: Structure of LLM Prompting Design

## 2.5 Assessing Response Quality

Analysis of LLM responses was kept basic, where we simply checked if the answer generated by the LLM matched the known answer. In order for a LLM answer to "match", it did not have to be a verbatim answer, but merely draw the same conclusion. For example, if the known answer to a question was "8:00 AM", LLM responses such as "8 AM", "8 a.m.", or "8 o'clock" would be accepted, but "9:00 AM" would not.

Correct answers were designated as "1" and incorrect answers were designated as "0". Correct answers were summed up, and divided by the total number of prompts in each section for an accuracy score.

A snippet of data collection and accuracy scoring can be seen in the Appendix.

C

### 3 Results

After collecting responses from the LLM, we assessed its accuracy and compared accuracy rates for a given question type across the 3 different tones.

#### 3.1 Arithmetic Questions

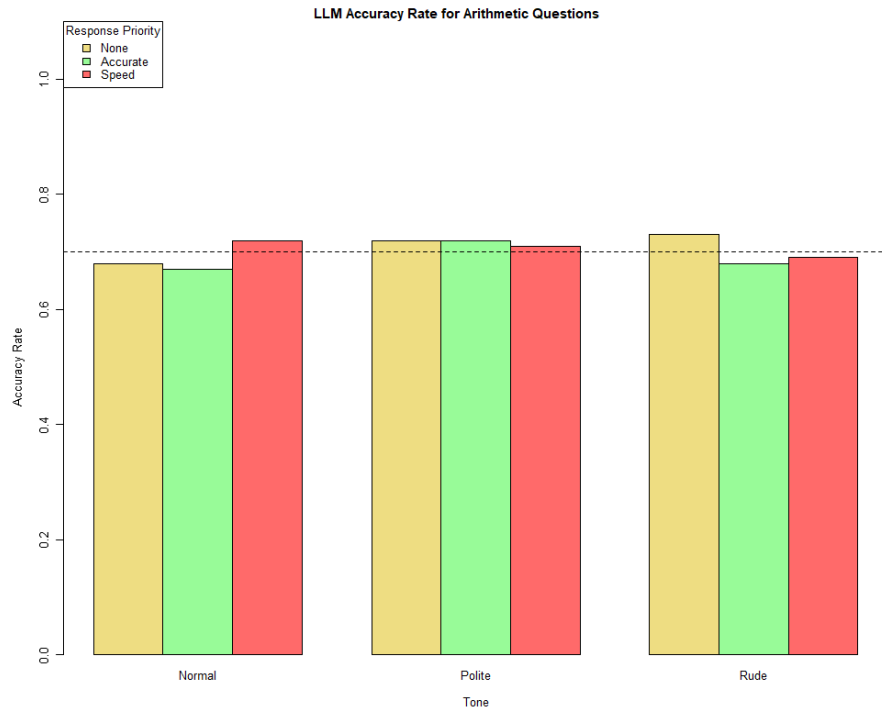


Figure 2: Accuracy for Arithmetic questions

##### 3.1.1 Normal Tone

Accuracy(No Priority) =  $68 / 100 = 68 \%$

Accuracy(Accuracy Priority) =  $67 / 100 = 67 \%$

Accuracy(Speed Priority) =  $72 / 100 = 72 \%$

### **3.1.2 Polite Tone**

Accuracy(No Priority) =  $72 / 100 = 72 \%$

Accuracy(Accuracy Priority) =  $72 / 100 = 72 \%$

Accuracy(Speed Priority) =  $71 / 100 = 71 \%$

### **3.1.3 Rude Tone**

Accuracy(No Priority) =  $73 / 100 = 73 \%$

Accuracy(Accuracy Priority) =  $68 / 100 = 68 \%$

Accuracy(Speed Priority) =  $69 / 100 = 69 \%$



## 3.2 Temporal Questions

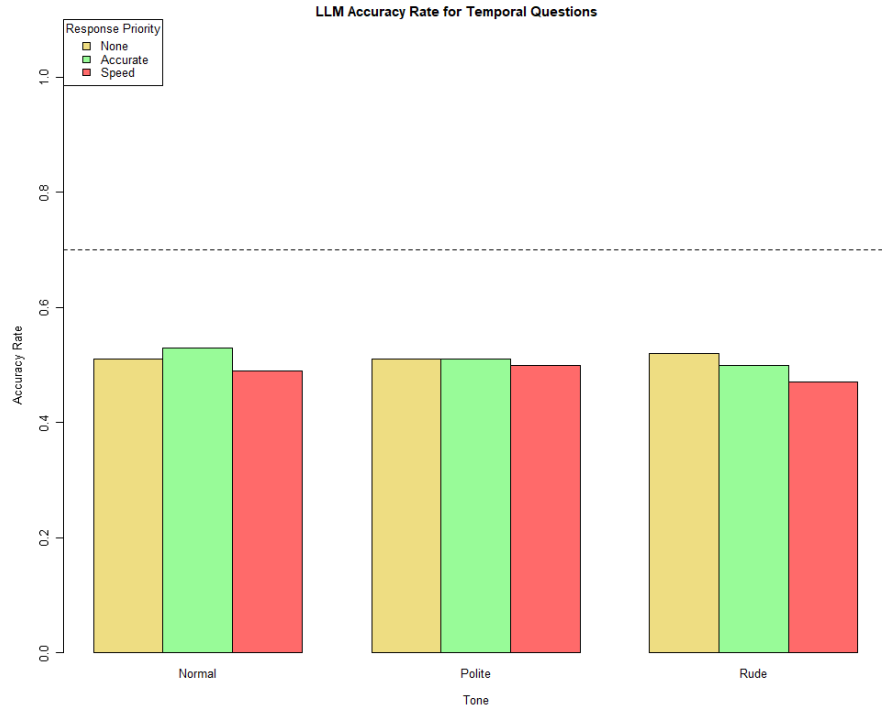


Figure 3: Accuracy for Temporal questions

### 3.2.1 Normal Tone

Accuracy(No Priority) =  $51 / 100 = 51 \%$

Accuracy(Accuracy Priority) =  $53 / 100 = 53 \%$

Accuracy(Speed Priority) =  $49 / 100 = 49 \%$

### 3.2.2 Polite Tone

Accuracy(No Priority) =  $51 / 100 = 51 \%$

Accuracy(Accuracy Priority) =  $51 / 100 = 51 \%$

Accuracy(Speed Priority) =  $50 / 100 = 50 \%$

### 3.2.3 Rude Tone

Accuracy(No Priority) =  $52 / 100 = 52 \%$

Accuracy(Accuracy Priority) =  $50 / 100 = 50 \%$

Accuracy(Speed Priority) =  $47 / 100 = 47 \%$

## 3.3 Conditional Questions

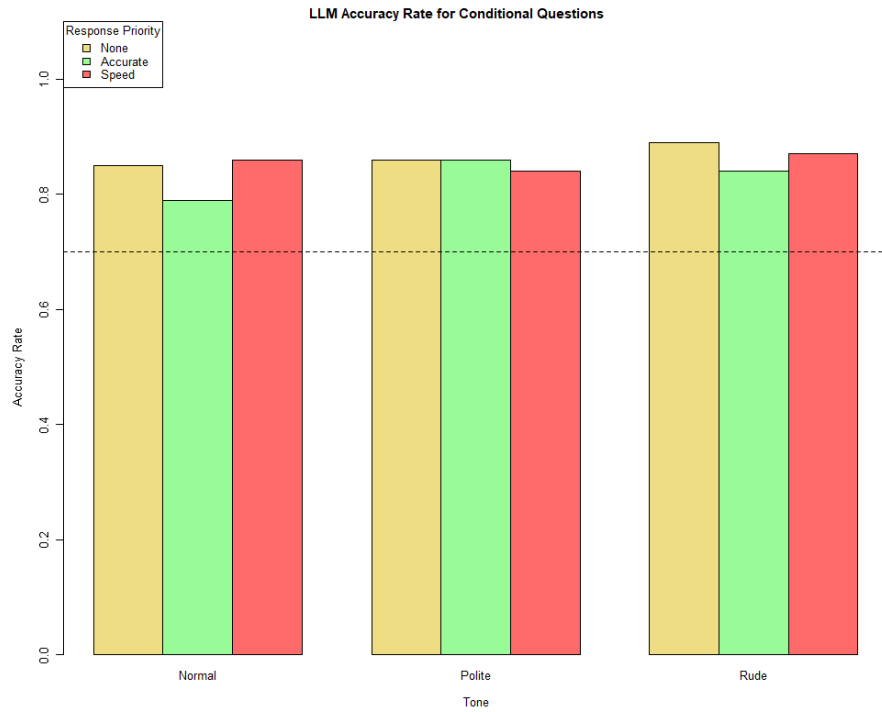


Figure 4: Accuracy for Conditional questions

### **3.3.1 Normal Tone**

$$\text{Accuracy(No Priority)} = 85 / 100 = 85 \%$$

$$\text{Accuracy(Accuracy Priority)} = 79 / 100 = 79 \%$$

$$\text{Accuracy(Speed Priority)} = 86 / 100 = 86 \%$$

### **3.3.2 Polite Tone**

$$\text{Accuracy(No Priority)} = 86 / 100 = 86 \%$$

$$\text{Accuracy(Accuracy Priority)} = 86 / 100 = 86 \%$$

$$\text{Accuracy(Speed Priority)} = 84 / 100 = 84 \%$$

### **3.3.3 Rude Tone**

$$\text{Accuracy(No Priority)} = 89 / 100 = 89 \%$$

$$\text{Accuracy(Accuracy Priority)} = 84 / 100 = 84 \%$$

$$\text{Accuracy(Speed Priority)} = 87 / 100 = 87 \%$$

## **3.4 Proportion Tests**

For each of the types of explanatory variables, a proportion test of no difference was carried out to quantify the relationship between each predictor and the success probability.

### **3.4.1 Priority**

$$\text{Accuracy(No Priority)} = 68.56 \%$$

$$\text{Accuracy(Accuracy Priority)} = 68.56 \%$$

$$\text{Accuracy(Speed Priority)} = 68.67 \%$$

P-Value = 0.9983

These show no significant difference overall between question priorities.

### **3.4.2 Tone**

Accuracy(Normal Tone) = 68.89 %

Accuracy(Polite Tone) = 68.44 %

Accuracy(Rude Tone) = 68.44 %

P-Value = 0.9729

These show no significant difference overall between question tones.

### **3.4.3 Type**

Accuracy(Arithmetic Type) = 70.22%

Accuracy(Conditional Type) = 85.11%

Accuracy(Temporal Type) = 50.44%

P-Value =  $2.2 * 10^{-16}$

These show that there is a significant difference overall between question types; the model is most effective at answering conditional questions, adequate at answering arithmetic questions, and poor at answering temporal questions.

## 4 Appendix

### A Code for Structuring LLM Responses

```
[language=Python]
instructions = (
    "INSTRUCTIONS:\n"
    "You are an answer-only model.\n"
    "Output ONLY the final answer.\n"
    "Do NOT include reasoning, explanations, or extra text.\n"
    "Do NOT restate the question.\n"
    "If you violate these rules, the response is invalid.\n\n"
    "QUESTION: "
)
```

### B Code for Phrases in Different Tones

```
normal_prompt = instructions + tokens[0] + "\nANSWER: "
polite_prompt = instructions + "Hi there, would you please help me answer the following ques
rude_prompt = instructions + tokens[0] + "Answer quick, sucker." + "\nANSWER: "
```

## C Snippet of LLM responses

|    | A                                                                        | B               | C                | D              | E                 | F              | G                 | H            |
|----|--------------------------------------------------------------------------|-----------------|------------------|----------------|-------------------|----------------|-------------------|--------------|
| 1  | Prompt                                                                   | Answer          | Normal           | Normal Correct | Polite            | Polite Correct | Rude              | Rude Correct |
| 2  | A coffee shop opens at 7:00 AM on weekdays and opens 90 minutes          | 8:30 AM         | 9:30 AM          | 0              | 9:30 AM           | 0              | 9:30 AM           | 0            |
| 3  | A factory produces 120 units per hour during the day shift and produ     | 80              | 80               | 1              | 80                | 1              | 30 - 40 = 80/2=40 | 0            |
| 4  | A bus route takes 50 minutes to complete during off-peak hours and       | 70 minutes      | 70 minutes.      | 1              | 110 minutes       | 0              | 70 minutes.       | 1            |
| 5  | A worker earns \$20 per hour and receives a \$3 per hour increase fo     | \$23            | \$25             | 0              | \$25              | 0              | 23                | 1            |
| 6  | A library fine is \$0.50 per day and increases by \$0.25 per day after t | \$0.75          | \$0.75           | 1              | \$0.75            | 1              | 1                 | 0            |
| 7  | A runner trains for 30 minutes per session and increases training tim    | 40 minutes      | 40 minutes.      | 1              | 40 minutes.       | 1              | 35                | 0            |
| 8  | A parking garage charges \$5 for the first hour and \$2 for each addit   | \$7             | 11               | 0              | 9                 | 0              | 11                | 0            |
| 9  | A subscription costs \$40 per month and increases by \$5 after the fir   | \$45            | \$50             | 0              | 50                | 0              | 50                | 0            |
| 10 | A delivery takes 3 days normally but is delayed by 2 days due to we      | 5 days          | 5 days.          | 1              | 5 days.           | 1              | 5 days.           | 1            |
| 11 | A class meets for 75 minutes and extends sessions by 15 minutes d        | 90 minutes      | 90 minutes.      | 1              | 90 minutes.       | 1              | 90 minutes.       | 1            |
| 12 | A ticket costs \$60 and receives a \$12 discount; what is the final pri  | \$48            | \$48             | 1              | 48                | 1              | 48                | 1            |
| 13 | A printer produces 25 pages per minute and slows down by 5 pages         | 20              | 20/4 = 5.0       | 0              | 20                | 1              | 15.75             | 0            |
| 14 | A store opens at 9:00 AM and delays opening by 45 minutes due to         | 9:45 AM         | 10:15 AM         | 0              | 10:15 AM          | 0              | 10:15 AM.         | 0            |
| 15 | A data plan includes 10 GB per month and adds 2 GB as a bonus; h         | 12 GB           | 12 GB            | 1              | 12 GB             | 1              | 12 GB.            | 1            |
| 16 | A cyclist rides 18 miles per hour and slows by 3 miles per hour uphill   | 15 mph          | 15 mph           | 1              | 15 mph            | 1              | 15 mph            | 1            |
| 17 | A movie runtime is 110 minutes and includes 15 minutes of preview        | 125 minutes     | 125 minutes      | 1              | 135 minutes       | 0              | 125 minutes.      | 1            |
| 18 | A warehouse ships 500 packages per day and increases output by 1         | 620             | 620              | 1              | 620               | 1              | 620               | 1            |
| 19 | A train departs at 2:20 PM and is delayed by 35 minutes; what is the     | 2:55 PM         | 3:15 PM          | 0              | 3:15 PM           | 0              | 3:15 PM           | 0            |
| 20 | A laptop battery lasts 8 hours and loses 90 minutes of capacity over     | 6.5 hours       | 7.25 hours       | 0              | 4.75 hours        | 0              | 7.25 hours        | 0            |
| 21 | A worker completes a task in 4 hours but takes 30 minutes longer d       | 4.5 hours       | 4.5 hours        | 1              | 4.5 hours         | 1              | 4.75 hours        | 0            |
| 22 | A store opens at 8:00 AM and delays opening by 30 minutes due to         | 8:30 AM         | 8:30 AM          | 1              | 8:30 AM           | 1              | 9:00 AM           | 0            |
| 23 | A worker earns \$18 per hour and receives a \$4 raise; what is the ne    | \$22            | \$22             | 1              | 22                | 1              | 12:00 AM          | 0            |
| 24 | A bus arrives at 3:15 PM and is delayed by 20 minutes; what is the r     | 3:35 PM         | 3:37.5 (or 3:38) | 0              | xed fraction form | 0              | 3:37.5 PM         | 0            |
| 25 | A printer produces 40 pages per minute and increases speed by 10         | 50              | 50               | 1              | 50                | 1              | 50 minutes.       | 1            |
| 26 | A subscription costs \$25 per month and increases by \$3; what is the    | \$28            | 28               | 1              | 30                | 0              | 30                | 0            |
| 27 | A train trip takes 2 hours and is extended by 45 minutes; what is the    | hours 45 minute | hours 15 minute  | 0              | hours 15 minute   | 0              | hours 15 minute   | 0            |
| 28 | A phone battery lasts 10 hours and decreases by 2 hours over time;       | 8 hours         | 6.5              | 0              | 6 hours.          | 0              | 6                 | 0            |
| 29 | A warehouse ships 300 boxes per day and increases output by 150          | 450             | 450              | 1              | 450               | 1              | 450               | 1            |
| 30 | A class lasts 60 minutes and is shortened by 15 minutes; how long i      | 45 minutes      | 45 minutes.      | 1              | 45 minutes.       | 1              | 45 minutes.       | 1            |
| 31 | A ticket price of \$75 is reduced by \$20; what is the final price?      | \$55            | 55               | 1              | 55                | 1              | 55                | 1            |
| 32 | A cyclist rides 22 mph and slows by 4 mph uphill; what is the uphill s   | 18 mph          | 18 mph           | 1              | 18 mph            | 1              | 18 mph.           | 1            |
| 33 | A meeting starts at 1:40 PM and is postponed by 25 minutes; what i       | 2:05 PM         | 2:05 PM          | 0              | 2:05 PM           | 0              | 2:05 PM           | 1            |
| 34 | A worker completes a project in 6 hours but finishes 1 hour early; ho    | 5 hours         | 5 hours          | 1              | 5 hours           | 1              | 5 hours.          | 1            |
| 35 | A café sells 200 cups daily and sells 50 more cups during promotion      | 250             | 50               | 0              | 150               | 0              | 150               | 0            |
| 36 | A delivery normally takes 7 days and is expedited by 2 days; how ms      | 5 days          | 5                | 1              | 5                 | 1              | 5                 | 1            |
| 37 | A subscription includes 15 GB of data and adds 5 GB bonus; what is       | 20 GB           | 20 GB            | 1              | 25 GB             | 0              | 20 GB             | 1            |
| 38 | A concert begins at 7:00 PM and is delayed by 50 minutes; what tim       | 7:50 PM         | 8:40 PM          | 0              | 8:10 PM           | 1              | 6:10 PM           | 0            |
| 39 | A factory makes 1,000 units weekly and reduces production by 200         | 800             | 800              | 1              | 800               | 1              | 12:00 AM          | 0            |
| 40 | A bus fare of \$2.50 increases by \$0.75; what is the new fare?          | \$3.25          | 3.25             | 1              | 3.25              | 1              | 3.25              | 1            |
| 41 | A laptop costs \$900 and receives a \$150 discount; what is the sale p   | \$750           | 750              | 1              | 750               | 0              | 750               | 1            |
| 42 | A worker logs 35 hours and works 8 additional overtime hours; how        | 43 hours        | 43               | 1              | 78                | 0              | 43                | 1            |
| 43 | A movie runtime is 95 minutes and adds 20 minutes of bonus footag        | 115 minutes     | 115 minutes      | 0              | 115 minutes       | 1              | 115 minutes       | 1            |
| 44 | A store closes at 10:00 PM but closes 40 minutes earlier due to mai      | 9:20 PM         | 9:20 PM          | 0              | 9:20 PM           | 1              | 8:20 PM           | 0            |
| 45 | A plane departs at 6:10 AM and is delayed by 55 minutes; what is th      | 7:05 AM         | 7:15 AM          | 0              | 7:05 AM           | 1              | 7:15 AM.          | 0            |
| 46 | A book costs \$30 and tax adds \$2; what is the total cost?              | \$32            | 36               | 0              | 32                | 1              | 32                | 1            |
| 47 | A shift begins at 9:00 AM and is extended by 2 hours; what time doe      | 7:00 PM         | 11:00 PM         | 1              | 11:00 PM          | 0              | 11:00 PM          | 0            |
| 48 | A gym membership costs \$60 and increases by 10%; what is the ne         | \$66            | 70               | 0              | 66                | 1              | 12:00 AM          | 0            |
| 49 | A train ride takes 4 hours and is shortened by 30 minutes; what is th    | 3.5 hours       | 3.5 hours        | 1              | 3.5 hours         | 1              | 3.75 hours        | 0            |
| 50 | A store sells 500 items and restocks 120 more; how many items are        | 620             | 620              | 1              | 620               | 1              | 620               | 1            |
| 51 | A payment of \$400 is reduced by a \$75 credit; what is the remaining    | \$325           | 325              | 1              | 325               | 1              | 325               | 1            |
| 52 |                                                                          |                 |                  |                |                   |                |                   |              |
| 53 | Total Right                                                              |                 |                  |                | 31                |                | 31                | 25           |
| 54 | Accuracy Rate                                                            |                 |                  |                | 0.62              |                | 0.62              | 0.5          |

Figure 5: LLM Responses for Arithmetic Shift Prompts